2023

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level C
Junior
Grades **7–8**

DATE

23 September

TIME ALLOWED

75 minutes

Instructions and Information 考试说明

1. You must maintain silence at all times.
   考试过程中请保持安静。
2. Do not open the paper until told to do so.
   收到监考老师指示前，请勿打开试卷。
3. Mobile phones are not allowed. If you have one you must give it to a supervisor now.
   禁止使用手机。如果携带手机，请立刻交给监考老师。
4. You must not leave your seat during the competition.
   考试期间禁止离开座位。
5. You may not borrow equipment without a supervisor's permission.
   未经监考老师许可，禁止向他人借用工具。
6. If you wish to leave the room a supervisor must accompany you.
   如果想离开考场，必须有监考老师陪同。
7. If you have any other questions, raise your hand and wait for a supervisor.
   如果有其他问题，请举手示意监考老师。
8. You are allowed a working time of 75 minutes. There is no extra reading time.
   本场考试时长为 75 分钟，没有额外的读题时间。
9. There are no penalties for incorrect answers. You should attempt all questions.
   答错不扣分，请尝试回答所有问题。
10. The questions have been thoroughly checked. Each question stands as written and no further explanation can be provided.
    本场考试的所有问题已仔细核查，不提供任何口头解释。
11. Diagrams are NOT drawn to scale. They are intended only as aids.
    图形未按比例绘制，仅供参考。
12. You may use printed language dictionaries.
    允许使用纸质语言词典。
13. You may NOT use calculators.
    不允许使用计算器

Important: No student may sit the competition more than once or sit more than one division of the competition. The AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

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Junior Division

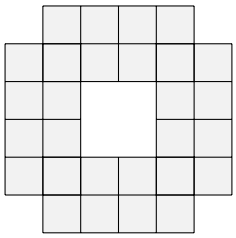
Questions 1 to 10 are worth 3 marks each.

1-10 题，每题 3 分

1. Kurt paved his courtyard in the pattern shown.

How many 1×1 pavers are in his courtyard?

Kurt 将庭院铺上如图所示的瓷砖。请问庭院中一共有多少块 1×1 的瓷砖？



(A) 28 (B) 30 (C) 32 (D) 34 (E) 36

---

2. Which of the following expressions has the smallest value?

请问以下哪个算式的值最小？

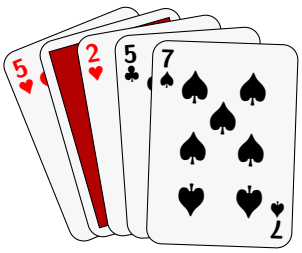
(A) $3 + 2$ (B) $3 - 2$ (C) 3×2 (D) $3 \div 2$ (E) 32

---

3. The numbers on the top corners of these five playing cards add to 21.

What is the number on the hidden card?

右图中五张扑克牌左上角的数字之和为 21。请问被遮住的扑克牌上的数字是多少？

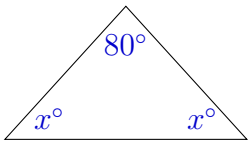


(A) 2 (B) 3 (C) 5 (D) 7 (E) 9

---

4. What is the value of x in this diagram?

请问图中 x 的值是多少？



(A) 50 (B) 60 (C) 70 (D) 80 (E) 100

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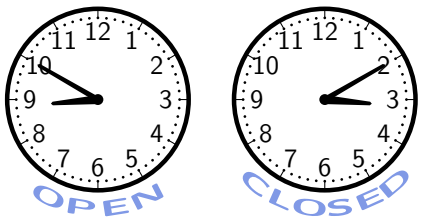
5. These two clocks show opening and closing times of my local Marine Discovery Centre. For how long is it open each day?
这两个时钟分别显示了当地海洋探索中心的开放（OPEN）和关闭(CLOSED)时间。请问这个中心每天会开放多长时间？



开放　　关闭

(A) 5 hours and 40 minutes　　5 小时 40 分钟

(B) 6 hours and 20 minutes　　6 小时 20 分钟

(C) 6 hours and 40 minutes　　6 小时 40 分钟

(D) 7 hours and 20 minutes　　7 小时 20 分钟

(E) 7 hours and 40 minutes　　7 小时 40 分钟

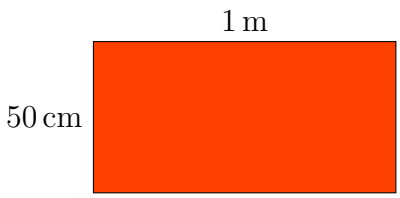
---

6. What is the value of $\dfrac{2+3+4}{7+8+9}$?
请问 $\dfrac{2+3+4}{7+8+9}$ 的值是多少？

(A) $\dfrac{1}{6}$　　(B) $\dfrac{2}{7}$　　(C) $\dfrac{3}{8}$　　(D) $\dfrac{4}{9}$　　(E) $\dfrac{1}{2}$

---

7. How many $25\,\text{cm} \times 25\,\text{cm}$ squares fit in a $50\,\text{cm} \times 1\,\text{m}$ rectangle?
请问有多少个 $25\,\text{cm} \times 25\,\text{cm}$ 的正方形可以完全放入 $50\,\text{cm} \times 1\,\text{m}$ 的长方形？



(A) 1　　(B) 2　　(C) 4　　(D) 6　　(E) 8

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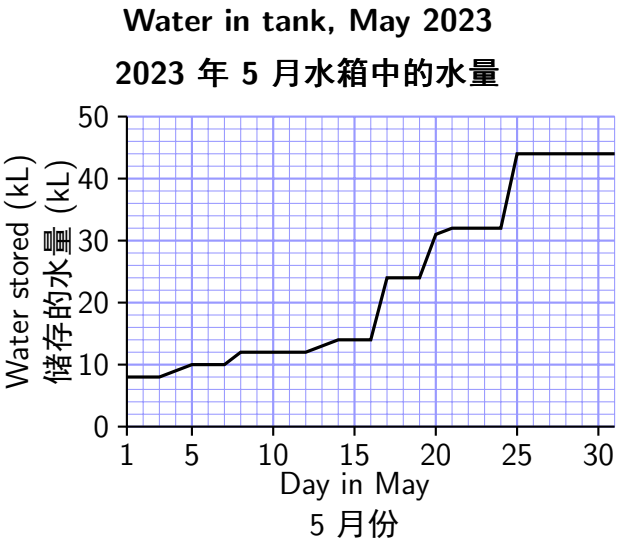
8. This graph shows the water stored in my rainwater tank last May.

What was the increase in water in the tank between the 10th and the 30th of May?

图中显示了今年 5 月份水箱中存储的水量情况。

请问 5 月 10 日到 30 日之间水箱中的水量增加了多少？

(A) 32 kL (B) 37 kL (C) 40 kL (D) 42 kL (E) 43 kL



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9. The sum of the ages of Ali, Bey and Cam is 32 years. What will be the sum of their ages in 3 years time?

Ali、Bey 和 Cam 三个人的年龄之和为 32 岁。请问 3 年后他们的年龄之和是多少？

(A) 33 years 33 岁 (B) 35 years 35 岁 (C) 38 years 38 岁 (D) 41 years 41 岁 (E) 96 years 96 岁

---

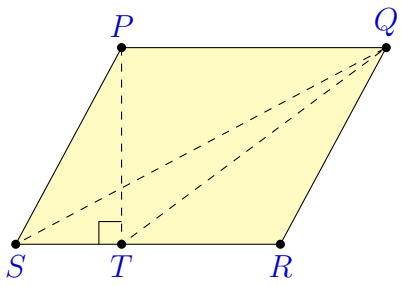
10. A parallelogram $PQRS$ has an area of $60\,\text{cm}^2$ and side PQ of length 10 cm.

Which length is 6 cm?

如图所示，平行四边形 $PQRS$ 的面积为 $60\,\text{cm}^2$，边 PQ 的长度为 10 cm。

请问哪条线段的长度是 6 cm？

(A) RQ (B) RS (C) QT (D) PT (E) QS



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Questions 11 to 20 are worth 4 marks each.
11-20 题，每题 4 分

11. Mei can travel to her grandma's house by a direct route, or by a scenic route that is 5 km longer. When she travels by the scenic route, and comes directly home, the round trip is 35 km. How long is the direct route?

小梅可以选择一条直达路线或者选择一条风景优美的路线去奶奶家，风景优美的路线比直达路线长 5 km。如果她选择风景优美的路线去奶奶家并选择直达路线回家，来回路程一共为 35 km。请问直达路线有多远？

(A) 5 km (B) 12.5 km (C) 15 km (D) 20 km (E) 22.5 km

---

12. In the game of Nurdle, players try to guess a three-digit number. After each guess, the digits are marked with one of the following three symbols:

5	3	5
×	⊙	×
1	2	3
×	✓	⊙

- ✓ if the digit is correct and in the right place
- ⊙ if the digit is correct, but is not yet in the right place
- × if the digit is not in the three-digit number at all.

Mike's guesses so far are shown.

How many different three-digit numbers fit the clues he has so far?

在 Nurdle 游戏中，玩家需要猜测一个三位数。每次猜测后，猜测的三位数的各位数字会用以下三种符号中的一种表示：

✓ 表示这个数字正确且位置正确

⊙ 表示这个数字正确，但是位置错误

× 表示这个数字不在这个三位数中

目前 Mike 的猜测如图所示。

请问有多少个不同的三位数符合目前已有的线索？

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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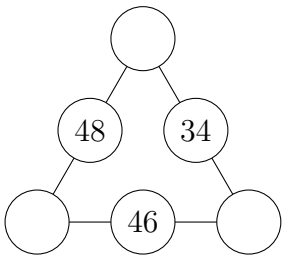
13. In this triangle, the number in the centre of each side is the sum of the two numbers at the ends of the sides.

What is the sum of the three numbers at the ends of the sides?

在这个三角形中，每条边中间的数字是两端数字之和。请问这三条边两端的三个数字之和是多少？



(A) 46 (B) 52 (C) 59 (D) 64 (E) 68

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14. A fuel tank is 40% empty. Then 40 litres of fuel is removed. The tank is now 40% full. How many litres are in a full tank?

一个油箱的 40% 是空的，取出 40 升油后，油箱的 40% 是满的。请问这个油箱可以装满多少升油？

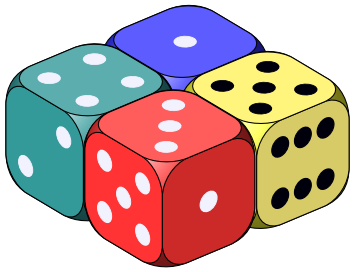
(A) 40 (B) 100 (C) 160 (D) 200 (E) 400

---

15. The game *Tutu* uses 4 standard dice with opposite faces adding to 7. The dice are rolled and randomly arranged into a 2×2 block.

A block's score is the sum of the 4 numbers showing on top plus the sum of the 8 numbers showing around the outside.

What is the largest possible score for the block shown?



游戏 *Tutu* 会用到 4 枚标准骰子，每枚骰子相对面的点数之和都为 7。投掷骰子并将骰子随机排列成 2×2 组合方块。组合方块的得分是顶面显示的 4 个点数与四周显示的 8 个点数之和。对于如图所示的组合方块，请问得分的最大可能值是多少？

(A) 44 (B) 46 (C) 48 (D) 50 (E) 52

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16. The five clocks below are all showing the wrong time. One clock is 3 minutes away from the correct time and another is 7 minutes away.

Which clock is closest to the correct time?

下面五个时钟显示的时间都是错误的。其中一个时钟与正确时间相差 3 分钟，另一个时钟与正确时间相差 7 分钟。请问哪个时钟最接近正确时间？

(A)

(B)

(C)

(D)

(E)

---

17. I wrote a list of all multiples of 7 between 1 and 500. Then I looked at the last digit of each number in my list.

Which digit appeared most often as the last digit?

我列出了 1 到 500 之间 7 的所有倍数，然后观察数列中每个数的最后一位数。请问哪项是最后一位数中出现频率最高的数字？

(A) 1 (B) 3 (C) 7 (D) 9

(E) None of (A)–(D), because there is no single last digit that appears most often
以上均不正确，因为最后一位数中没有出现频率最高的数字

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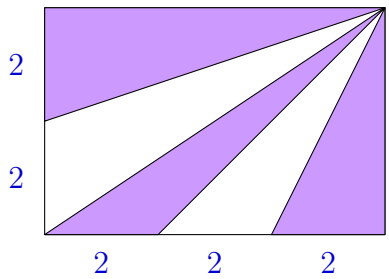
18. Amy designed this rectangular flag for her fleet of yachts. What fraction of the flag is shaded?

Amy 为游艇舰队设计了一面长方形旗帜。请问阴影部分的面积占旗帜总面积的几分之几？



(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{5}{8}$

(D) $\frac{1}{2}$ (E) $\frac{7}{12}$

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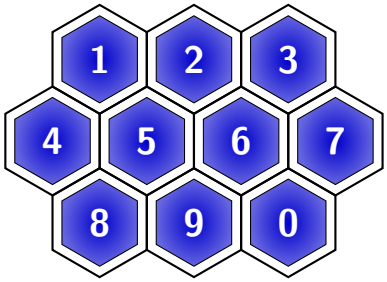
19. To unlock a security door, Daniyal enters a code on the keypad shown. To make it easy to remember:



- the code has 10 digits
- the code uses every digit exactly once
- the code steps from button to button only if the two buttons touch
- the code is as large as possible.

What is the 5th digit of Daniyal's code?

为了打开一扇安全门，Daniyal 在如图所示的键盘上输入一串密码。
为了方便记忆：

- 这个密码由 10 个数字组成
- 这个密码使用每个数字恰好一次
- 这个密码只能从一个按键上的数字到另一个相邻按键上的数字
- 这个密码的数值尽可能大

请问这个密码的第 5 个数字是多少？

(A) 0 (B) 1 (C) 2 (D) 6 (E) 7

---

20. Jake had 3 times as many gold balloons as blue balloons, and no other colours. Then some blew away, leaving him with 5 times as many blue balloons as gold balloons.

What is the smallest number of balloons that Jake could have started with?

Jake 拥有的金色气球数量是蓝色气球的 3 倍，且没有其他颜色的气球。然后有一些气球被吹走了，剩下的气球中，蓝色气球是金色气球的 5 倍。

请问 Jake 刚开始最少有多少个气球？

(A) 15 (B) 20 (C) 24 (D) 30 (E) 100

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Questions 21 to 25 are worth 5 marks each.

21-25 题，每题 5 分

21. Li attempted to multiply a single-digit number by 36, but he accidentally multiplied by 63 instead. His answer was 189 larger than the correct answer.

What was the correct answer to the multiplication?

Li 尝试将一个一位数乘以 36，但他不小心将这个一位数乘以了 63。最终得到的结果比正确结果大 189。请问这个乘法计算的正确结果是多少？

(A) 180 (B) 189 (C) 216 (D) 252 (E) 288

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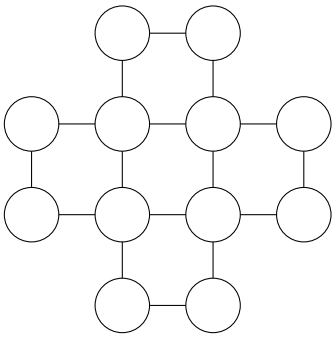
22. The numbers from 1 to 12 are to be placed on the vertices of the 5 interconnected squares in this diagram. The sum of the 4 vertices of any of the 5 squares is a constant. What is this constant?



将数字 1 到 12 放置在右图中 5 个相互连接的正方形的顶点上。每个正方形的 4 个顶点上的数字之和都是一个常数。

请问这个常数是多少？

(A) 26 (B) 28 (C) 30 (D) 32 (E) 34

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23. Ann is running round her favourite running course at a steady pace of 5 minutes per kilometre. She passes a marker which she knows is one-quarter of the way round. One minute and 40 seconds later, she passes another marker which is one-third of the way round. How long will it take her to run the whole course?

Ann 正在绕着她最喜欢的跑道以 5 分钟/公里匀速**前进**，她经过了一个标记牌，表示跑道的四分之一。一分钟四十秒后，她经过了另一个标记牌，表示跑道的三分之一。请问她跑完整个跑道需要多长时间？

(A) 18 minutes 18 分钟

(B) 20 minutes 20 分钟

(C) 20 minutes and 40 seconds 20 分钟 40 秒

(D) 22 minutes and 30 seconds 22 分钟 30 秒

(E) 24 minutes 24 分钟

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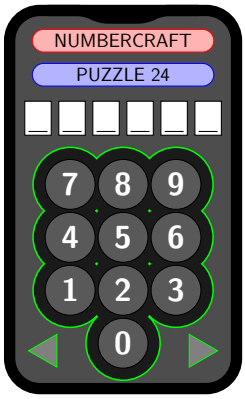
24. Jaz unlocked a secret level in the game Numbercraft. She entered a 6-digit code where the sum of the first three digits equals the sum of the last three digits. Further, she gained a bonus, since her code had four 7s in a row.

For instance, she could have entered 077770.

How many different codes could she have entered to get the secret level and the bonus?



Jaz 解锁了 Numbercraft 游戏的一个秘密等级。她输入了一个六位数密码，其中前三位数字之和等于后三位数字之和。此外，因为她的密码中有四个连续的数字 7，她还得到了额外奖励。例如，她输入的密码可能是 077770。

为了解锁秘密等级和获得奖励，请问她可以输入多少个不同的密码？

(A) 10 (B) 18 (C) 20 (D) 27 (E) 64

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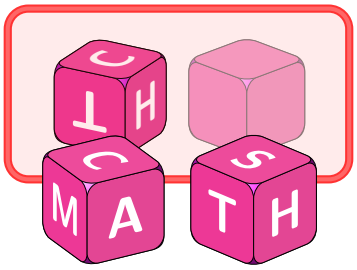
25. Two identical dice have faces labelled A, C, H, M, S and T.

The diagram shows what they look like when placed in front of a mirror, but the reflection of the second dice has been left blank.

What does the reflection of the second dice look like?



两枚相同骰子的六个面上分别标记有 A、C、H、M、S 和 T。图中展示了将它们放在镜子前呈现的图像，但第二枚骰子的图像是空白的。

请问下列哪项是第二枚骰子在镜子中的图像？

(A)
(B)
(C)
(D)
(E)

For questions 26 to 30, colour in the bubbles on the answer sheet to record whole-number answers from 0 to 999. Questions 26 to 30 are worth 6, 7, 8, 9 and 10 marks respectively.

26 题至 30 题的答案为 0 至 999 的整数，请正确填涂在答题卡上。第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，第 29 题占 9 分，第 30 题占 10 分。

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26. I start with a number, multiply it by 10, and then subtract a multiple of 9 that is less than 100. My answer is 5347. What number did I start with?

我先将某个数字乘以 10，然后减去一个小于 100 的 9 的倍数，得到的最终结果是 5347。请问起始数字是多少？

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27. Amelia noticed that the names of three friends, Mei, Emma and Liam, were all made from letters of 'Amelia'. She chose different values from 0 to 9 for each of A, E, I, L and M so that

$$M + E + I \;=\; E + M + M + A \;=\; L + I + A + M$$

What is the largest possible value of $A + M + E + L + I + A$?

Amelia 注意到三个朋友 Mei、Emma 和 Liam 的名字都是由“Amelia”中的字母组成的。她给 A、 E、I、L 和 M 中的每个字母选定了从 0 到 9 之间不同的值，使得

$$M + E + I \;=\; E + M + M + A \;=\; L + I + A + M$$

请问 $A + M + E + L + I + A$ 的最大可能值是多少？

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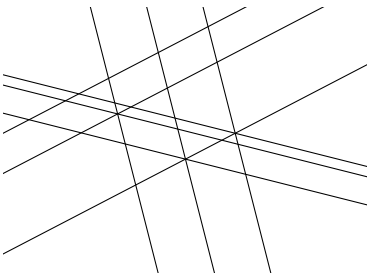
28. There are three sets of three parallel lines in a plane. Lines in different sets are not parallel, and every pair of non-parallel lines intersect. The diagram shows one way to do this, but for other arrangements the number of points of intersection varies.



Find the largest possible number of intersection points and the smallest possible number of intersection points, then multiply these two numbers.

一个平面上有三组平行线，每组各三条平行线。不同组中的线是不平行的，每对非平行线会相交。图中显示了一种可能的方式，但对于其他排列方式，交点的数量各不相同。

请问交点数量的最大可能值和最小可能值之积是多少？

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29. A two-digit number has the property that when it is divided by the sum of its digits the result is 4 with remainder 3. What is the sum of all two-digit numbers with this property?

一个两位数具有以下性质：当这个两位数除以它的各位数字之和时，商为 4，余数为 3。请问具有这个性质的所有两位数之和是多少？

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30. Donna likes all numbers that are not divisible by 3 and Sandra likes all numbers that have no digits divisible by 3. How many four-digit numbers are there that both Donna and Sandra like?

Donna 喜欢不能被 3 整除的所有数字，Sandra 喜欢各个数位上的数字都不能被 3 整除的所有数字。请问有多少个四位数是 Donna 和 Sandra 都喜欢的数字？

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